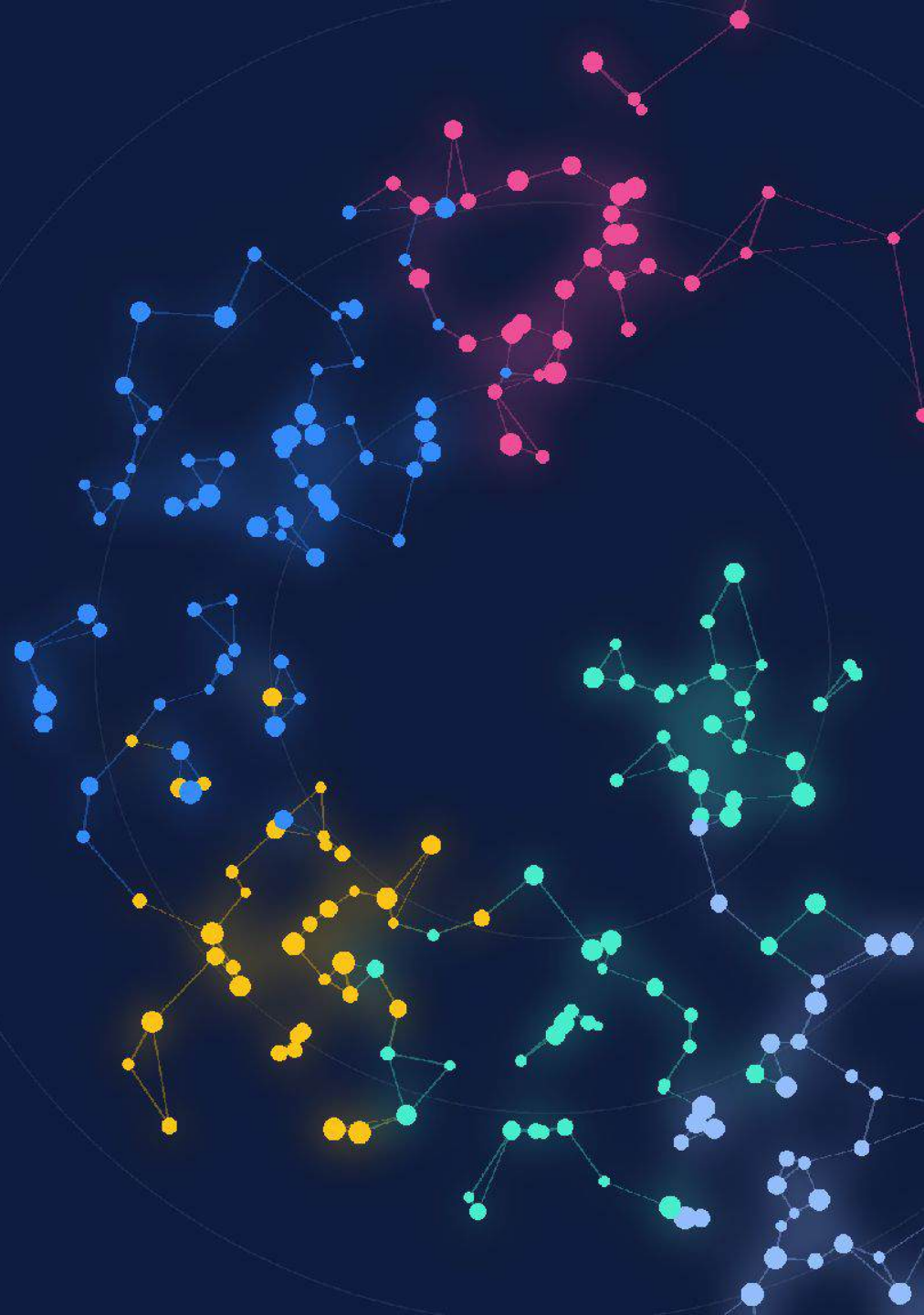
PULUMI · ELASTIC

Building AI Applications with Jina Models

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Principal Product Manager

elastic



ABOUT ME

Hi, I'm Kapil



Principal Product Manager

12 years in Search, AI and Voice Assistants

Microsoft

bloomreach

Connect with me

elastic

TRIVIA TIME

When was the Transformer paper published?

“Attention Is All You Need” changed how we build language and search models. Which year did it appear?

A

2014

B

2017

C

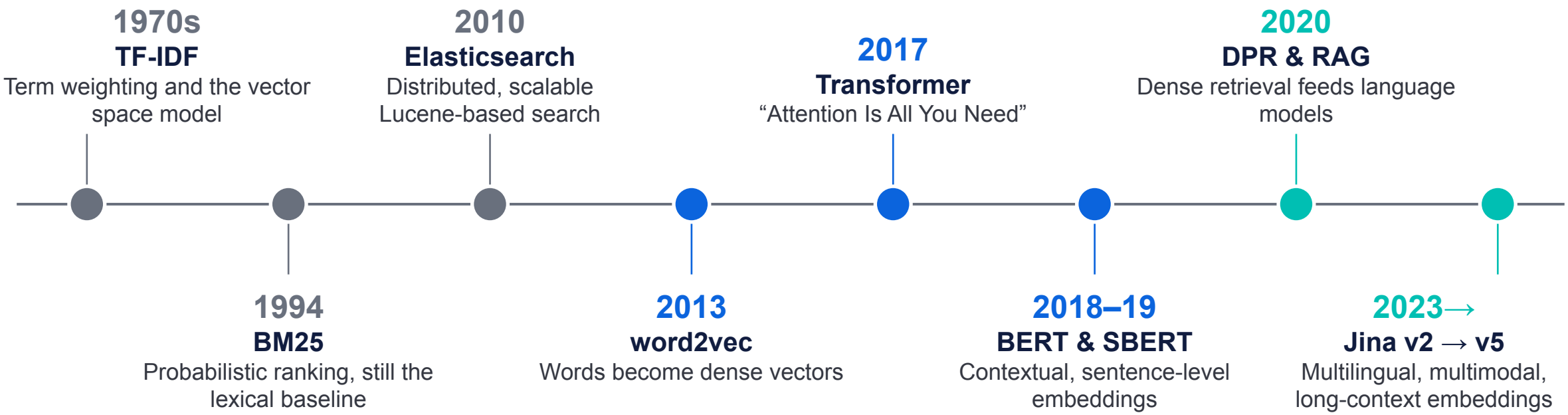
2019

D

2021

Raise your hand: A, B, C or D?

50 years of finding things



LEXICAL · match the words

NEURAL · match the meaning

RETRIEVAL FOR LLMs · match the context

Keywords match words, not meaning

my laptop won't turn on

KEYWORD SEARCH (BM25)

- **Turn on** notifications on your **laptop**
  Shares the words, wrong intent
- How to **turn on** dark mode in apps
  Shares the words, wrong intent

Ranks by overlapping terms

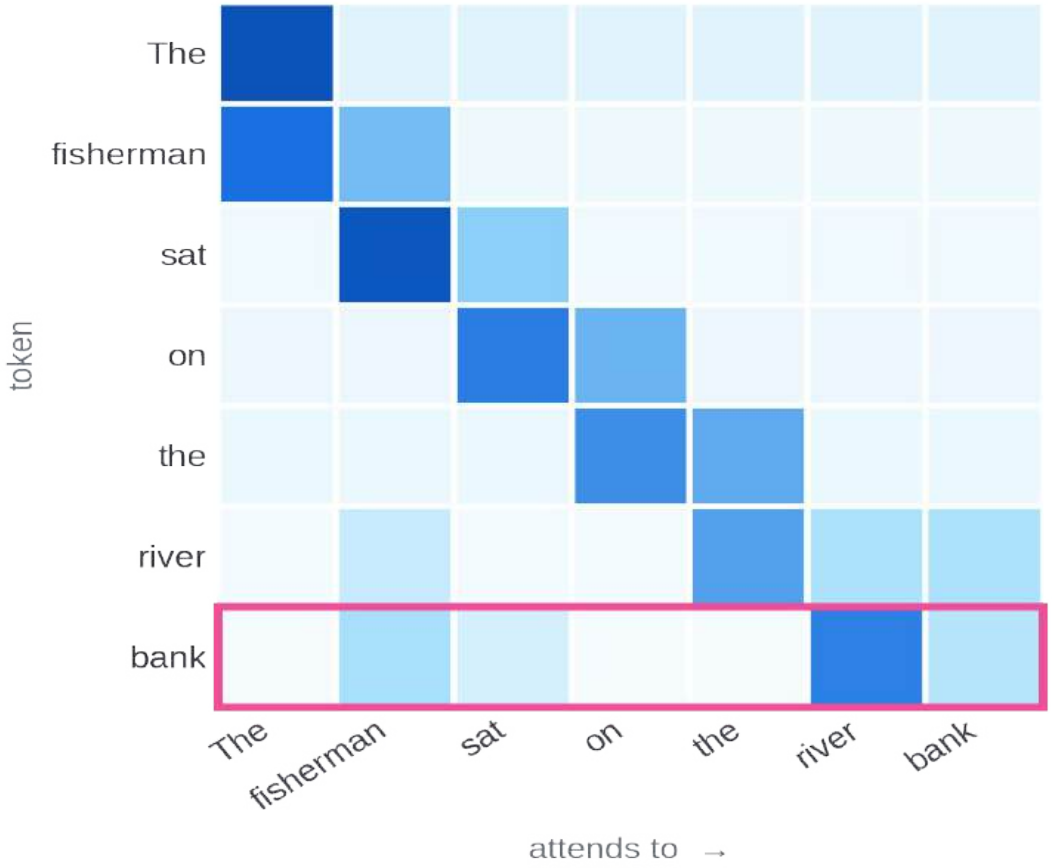
SEMANTIC SEARCH (EMBEDDINGS)

- Troubleshooting: device fails to power up
  No shared words, same meaning
- Battery drained? Charging & power-button fixes
  No shared words, same meaning

Ranks by closeness in meaning

Illustrative example. Pink words are the terms a keyword engine matches on.

Transformers: context for every token



Illustrative attention pattern: “bank” leans on “river”

Self-attention

Every token weighs every other token, so “bank” near “river” means something different than near “loan”.

Parallel and scalable

Trains on web-scale data in parallel, then adapts to downstream tasks like retrieval.

Two families

Encoders (BERT, EuroBERT) read in both directions. Decoders (GPT, Qwen3) predict the next token. Both now power embedders.

Embeddings: meaning as geometry

1

Encode

A transformer turns your text into contextual token states.

2

Pool

Collapse them into one fixed-size vector, e.g. 1,024 numbers.

3

Train contrastively

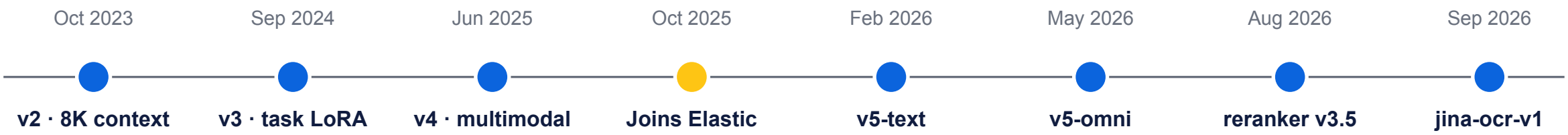
Pull matching query–document pairs together, push everything else apart.

Search = nearest neighbours by cosine similarity



A 2-D sketch of a 1,024-D space: similar meaning sits close together

The Jina search foundation family



Text embeddings

jina-embeddings-v5-text

small 677M · nano 239M. Multilingual, up to 32K context.

Omni embeddings

jina-embeddings-v5-omni

Text, image, audio, video and PDF in one vector space.

Rerankers

jina-reranker-v3.5

0.6B listwise reranker. jina-reranker-m0 for visual docs.

Parsing & Reader

jina-ocr-v1 · Reader

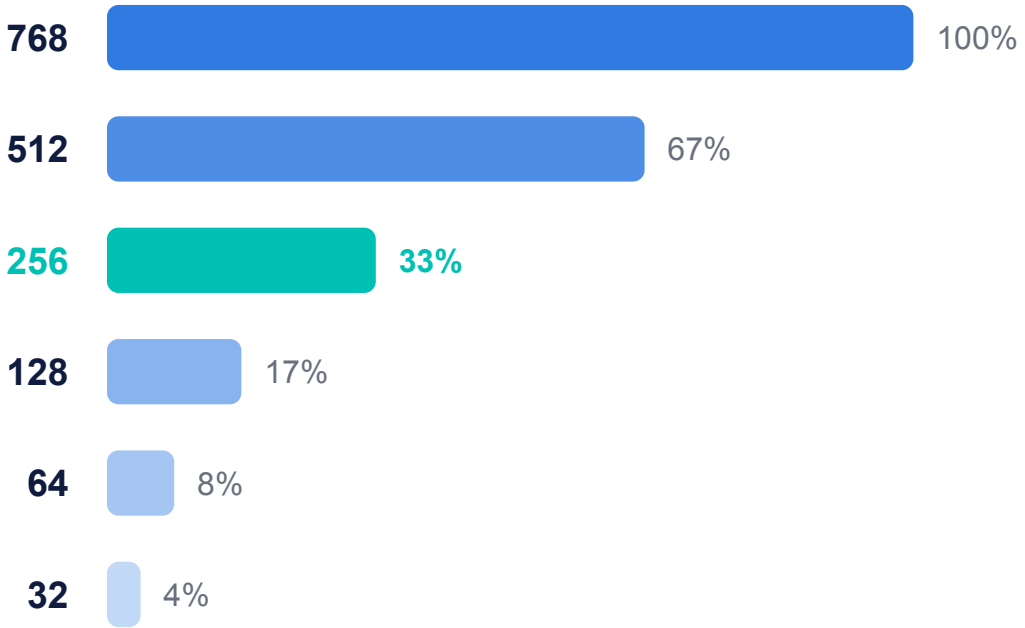
Pages, PDFs and URLs to clean Markdown for LLMs.

Open weights on Hugging Face (CC BY-NC 4.0). Commercial use via Elastic Inference Service, Jina API or Jina On-Prem.

Inside jina-embeddings-v5-text

Matryoshka: truncate to fit your budget

v5-text-nano, relative vector size



256 dims keeps >95% of retrieval quality at ⅓ the storage

Distilled from a 4B teacher

Qwen3-Embedding-4B quality compressed into 677M (small) and 239M (nano) students.

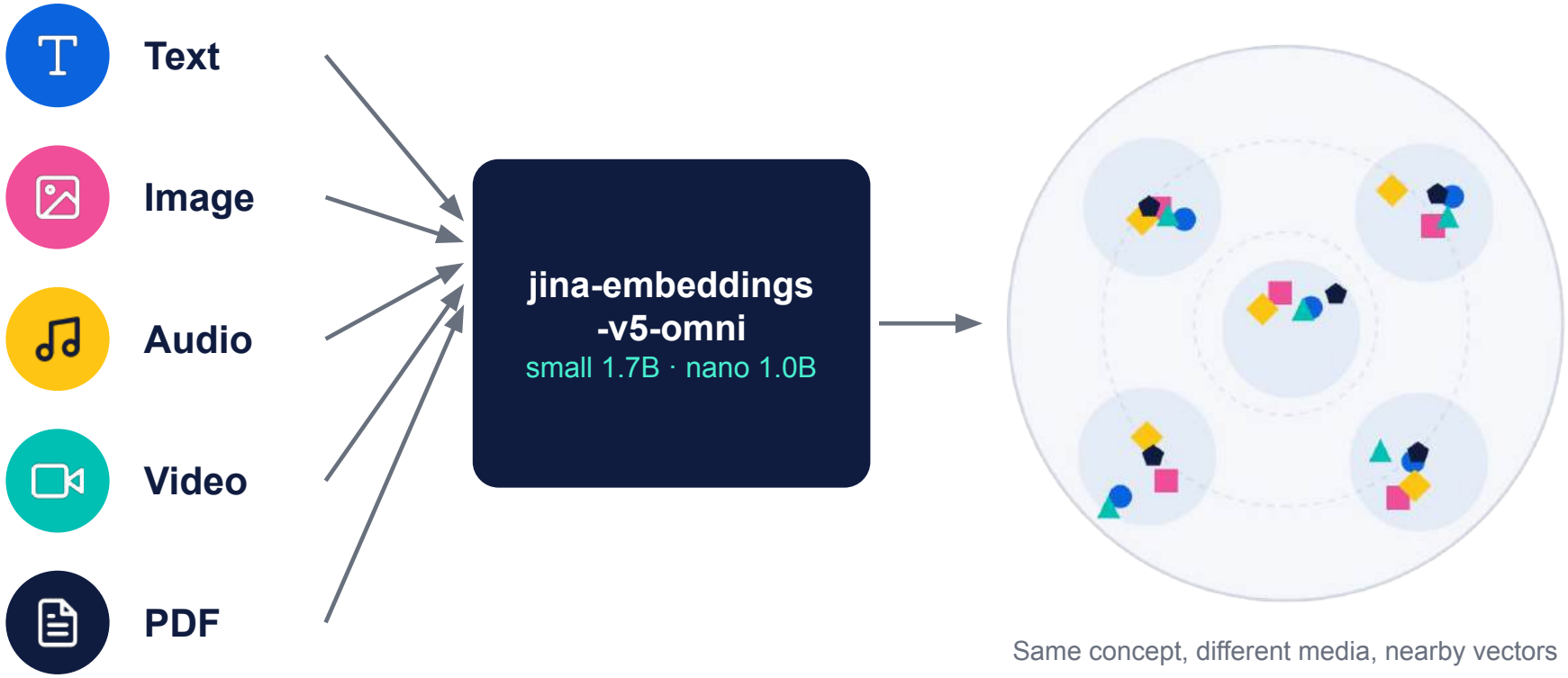
Four task LoRA adapters

Retrieval, text-matching, clustering, classification. Pick one per request, at negligible overhead.

Quantization-friendly

GOR regularization keeps binary embeddings nearly lossless. GGUF and MLX builds run on CPU and Apple silicon.

v5-omni: one space, every modality



Same concept, different media, nearby vectors

0.35%

of weights trained; text and media encoders stay frozen

0

re-embeds needed: text vectors are identical to v5-text

5.7×

fewer params than LCO-7B at a similar 4-modality average

Search images, audio and video with a text query, in the same Elasticsearch index

Rerankers: precision where it counts

Retrieve

Top 100–1,000 · BM25 + vectors · milliseconds

Rerank

Top 50–100 → jina-reranker-v3.5

Generate

Top 5–10 → your LLM or agent

Pointwise (cross-encoder)

query + doc 1 | query + doc 2 | query + doc 3

Each document is scored in isolation

Listwise: “last but not late” (Jina)

query + doc 1 + doc 2 + doc 3 + … (one 131K-token context)

Candidates are compared against each other before scoring. v3.5 adds hybrid attention (sliding-window + global layers) for up to 1.56× faster inference.

What you can build

Hybrid semantic search

Blend BM25 and vectors across product, docs and support content.

v5-text-small

RAG and chat over your data

Ground LLM answers in retrieved, reranked context.

v5-text + reranker v3.5

Multimodal search

Find images, audio, video and PDF pages with a text query.

v5-omni

Global, multilingual apps

One model, 100+ languages, cross-lingual retrieval.

v5-text / v5-omni

Agent memory and context

Low-latency retrieval of tools, memories and docs for agents.

v5-text-nano

Classify, cluster, dedupe

Task adapters for text-matching, clustering and classification.

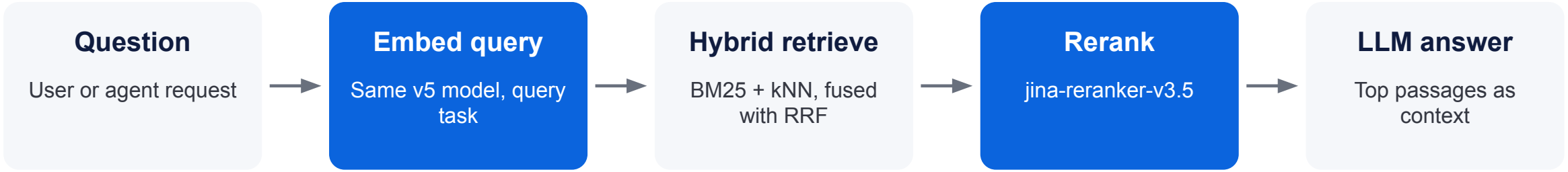
task LoRA adapters

Reference architecture for RAG

INGEST · OFFLINE



QUERY · ONLINE



Jina model

Run the Jina steps on Elastic Inference Service, the Jina API, or self-hosted.

Ship it in a few lines

1 · EMBED + INDEX WITH semantic_text

```
PUT _inference/text_embedding/jina-v5
{ "service": "elastic",
  "service_settings": {
    "model_id": "jina-embeddings-v5-text-small" } }

PUT docs
{ "mappings": { "properties": {
  "content": { "type": "semantic_text",
               "inference_id": "jina-v5" } } } }

POST docs/_doc
{ "content": "Reset your SSO password in Settings." }
```

2 · SEARCH + RERANK WITH A RETRIEVER

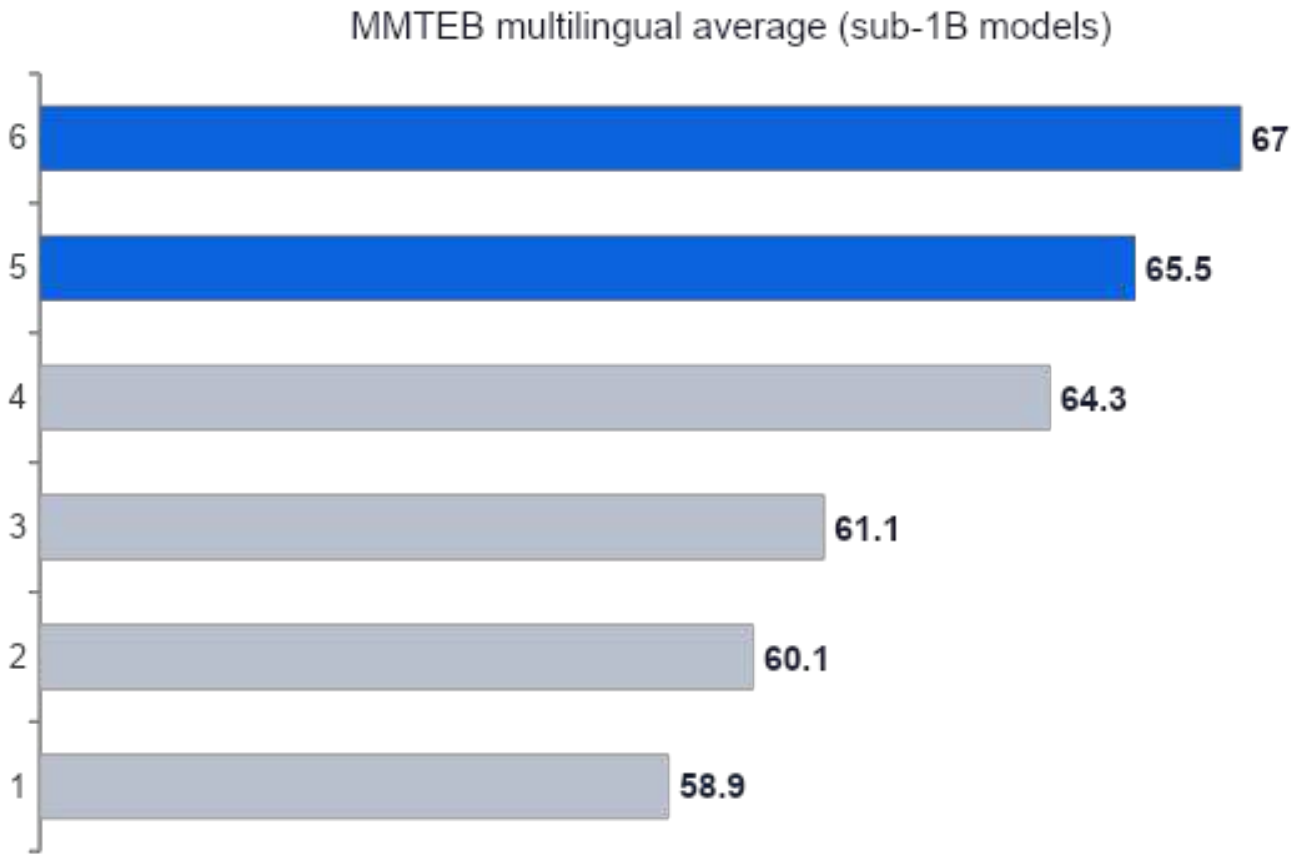
```
PUT _inference/rerank/jina-rerank
{ "service": "elastic",
  "service_settings": {
    "model_id": "jina-reranker-v3.5" } }

GET docs/_search
{ "retriever": { "text_similarity_reranker": {
  "retriever": { "standard": { "query": {
    "semantic": { "field": "content",
      "query": "how do I reset my SSO?" } } } },
  "field": "content",
  "inference_id": "jina-rerank",
  "inference_text": "how do I reset my SSO?",
  "rank_window_size": 50 } } }
```

Run it your way

Elastic Inference Service · Jina API · Hugging Face · vLLM · llama.cpp · MLX · AWS SageMaker · Jina On-Prem

Embeddings: small models, big scores



67.0

MMTEB score for v5-text-small, +2.7 over the next-best sub-1B model

5.6×

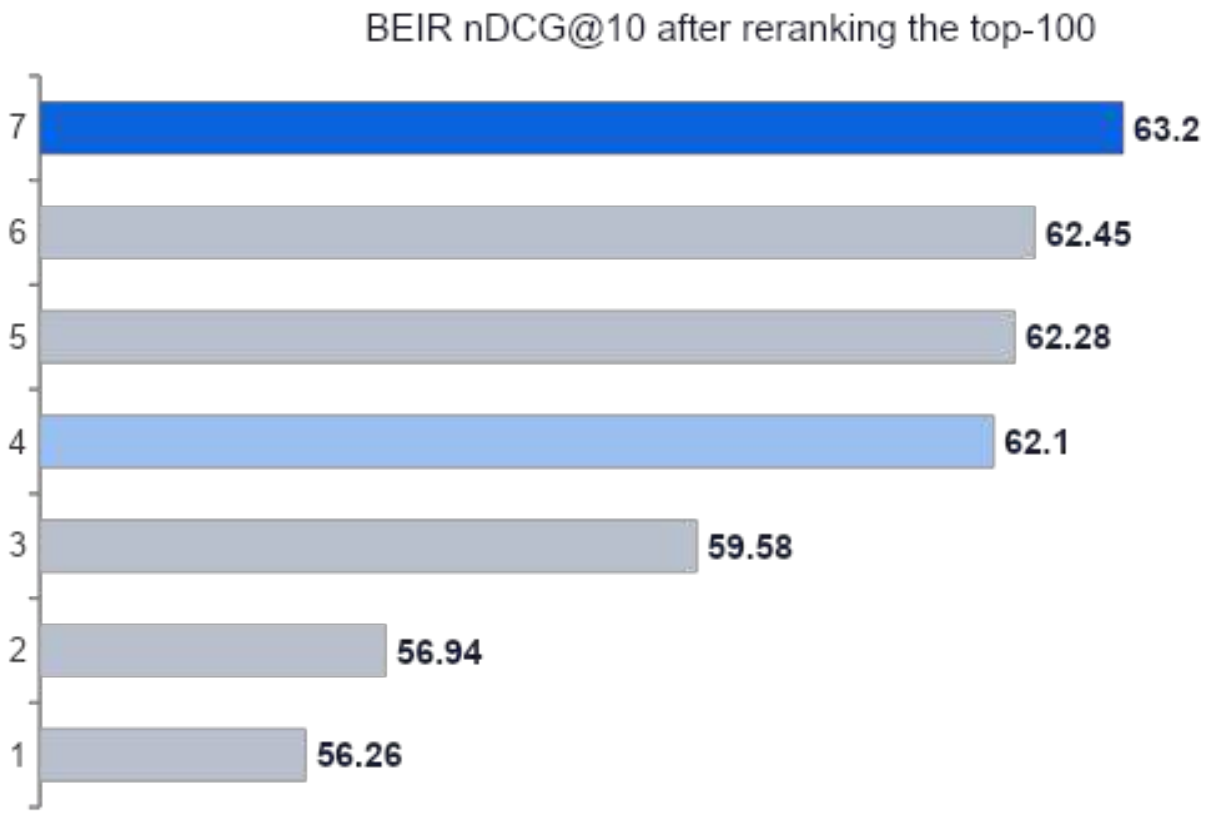
smaller than jina-embeddings-v4 (3.8B) at matching retrieval quality: 63.28 vs 63.62

71.7

MTEB English average, leading sub-1B multilingual models

Source: Jina AI, Feb 2026. MMTEB task-level mean over 131 tasks. Axis starts at 50.

Rerankers: 0.6B that rivals 4B



63.20

BEIR nDCG@10, ahead of the 4B Qwen3 reranker at ~7× fewer parameters

1.56×

faster than v3 on long documents (16.1 s → 10.3 s per query)

+9.6

nDCG@10 over v3 on semi-structured, field-constrained retrieval

Fair note: Qwen3-Reranker-4B still leads on MIRACL (76.56 vs 74.11) and RTEB (77.68 vs 70.95).

Source: Jina AI, Aug 2026. BEIR over 13 datasets, reranking top-100 from v5-text-small. Latency on one A100, batch size 1. Axis starts at 54.

Choosing the right model

If you need…	Start with	Key specs
Edge, CPU or latency-sensitive apps	**jina-embeddings-v5-text-nano**	239M · 8K ctx · 768 dims
Best text accuracy, long documents	**jina-embeddings-v5-text-small**	677M · 32K ctx · 1024 dims
Images, audio, video and PDFs in one index	**jina-embeddings-v5-omni**	small 1.7B / nano 1.0B
Sharper top results, structured data	**jina-reranker-v3.5**	0.6B · listwise · 131K ctx
Ranking visual documents	**jina-reranker-m0**	Multimodal reranker
PDFs, pages and URLs to Markdown	**jina-ocr-v1 · Reader**	3.4B total, 570M active

Open weights are CC BY-NC 4.0. Commercial production use via Elastic Inference Service, Jina API or Jina On-Prem.

TAKEAWAYS

Three things to remember

01

Retrieval is the ceiling

An AI app is only as good as the context it retrieves. Embeddings plus reranking is the highest-leverage upgrade.

02

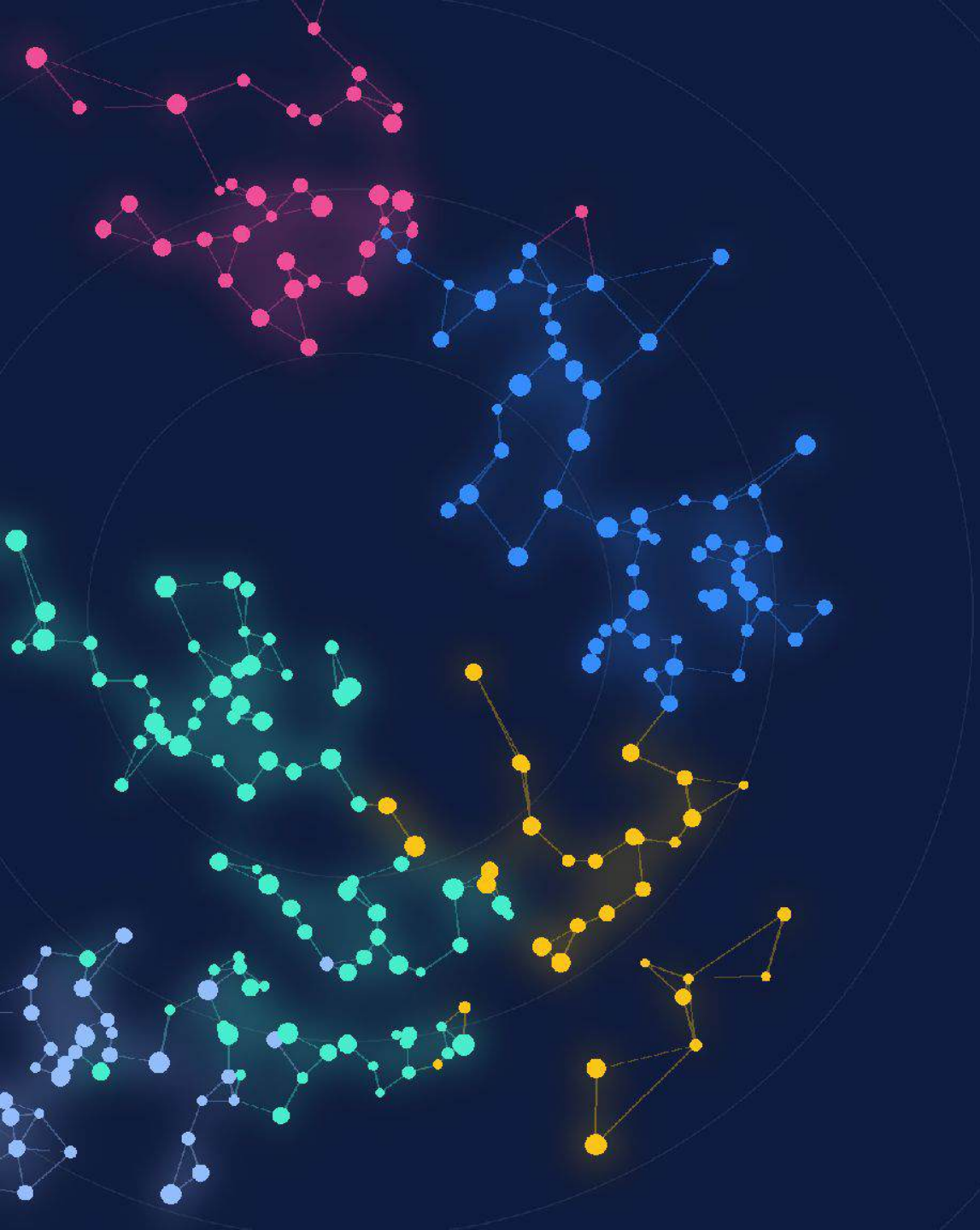
Small is the new state of the art

Sub-1B models now match multi-billion-parameter ones: cheaper, faster, deployable anywhere.

03

One stack, any deployment

Elastic Inference Service, Jina API, open weights, or fully air-gapped on-prem.



Questions?

Thank you. Let's talk retrieval.

- **jina.ai/models**
- **elastic.co/search-labs**
- **huggingface.co/jinaai**